

Mid Semester Examination- 2025

Name of the Program: B. Tech

Semester: vi

Name of the Course: Full Stack Web Development

Course Code: **TCS 693**

Time: 1-1/2 Hour

Maximum Marks: 50

Note:

- (i) Answer **all the questions** by choosing **any one of the sub questions**.
- (ii) Each question carries 10 marks

Q1	(10 marks)	
(a)	Define Meta tags with suitable properties. Also explain some suitable tags of HTML5.	CO1, CO2
OR		
(b)	What do you understand by the front end and back-end Technologies? Also, Create an HTML webpage that implements an image map . Use an image of a world map , and define clickable areas for at least three countries . Each area should redirect the user to a relevant webpage or display additional information when clicked.	
Q2	(10 marks)	
(a)	Explain document object model (DOM)? Explain some of the methods of array object. Write a program to manipulate DOM by changing background color of a division with random colors on every click of a button.	CO2,CO4
OR		
(b)	Demonstrate the practical implementation of <frame> and <iframe> in HTML. Explain their differences and use cases with appropriate examples. Additionally, create an HTML webpage that uses an iframe to embed a YouTube video and a separate frame layout to display two different webpages side by side.	
Q3	(10 marks)	
(a)	Explain following attributes and input elements of an HTML form using suitable examples – type, name, placeholder, method, action, value, required. Also, validate all input controls with the help of JavaScript. For ex. Length of phone, no should be 10 digits, Email is of proper format etc. HTML Forms First name: Last name: Age: ☒ Male ☐ Female Submit Reset	CO2,CO3

	OR	
(b)	Differentiate between the GET and POST methods in form handling, explaining their use cases and security implications. How can you apply a @keyframes animation to an element in CSS? Explain the purpose of from and to in the @keyframes rule.	
Q4	(10 marks)	CO3, CO4
(a)	Explain the Box Model in CSS, including its components and how it affects the layout of elements on a webpage. Additionally, differentiate between Pseudo-classes and Pseudo-elements in CSS with suitable examples demonstrating their usage.	
	OR	
(b)	Explain split, search, and replace functions of Javascript along with an example? Use random function to generate numbers in a range low – high, where low and high will be taken as input from the user.	
Q5	(10 marks)	CO4
(a)	Explain the use of JavaScript functions like <code>createElement()</code> , <code>querySelector()</code> and <code>addEventListener()</code> . Write a program that dynamically adds buttons to the document using an event handler in JavaScript.	
	OR	
(b)	Explain the concept of backend data loading. Additionally, describe various getter and setter methods of the JavaScript Date object with examples. Furthermore, write a JavaScript program that takes a string as input and returns a new string with all the vowels removed.	

Note for the question paper setters: (Assuming two units are covered)

- Question paper should have questions from both the units covering the related COs
- Sub questions (a) and (b) can be further divided if required